

Priyanka Singh

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Education

Vishwakarma Institute of Technology

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI & ML) | CGPA: 9.32 / 10

Pune, Maharashtra

June 2028

Kendriya Vidyalaya No. 2

Senior Secondary (Class XII): 92% | Secondary (Class X): 94.8%

Pune, Maharashtra

2023

Projects

PCOD Detection using Machine Learning | [GitHub](#)

- Built a data pipeline merging two clinical datasets (541×44, 1000×6 records) via KNN imputation, z-score normalization, and SMOTE to resolve missing values, inconsistency, and class imbalance
- Applied 5 feature selection techniques (SHAP, Boruta, Lasso, RFECV, mRMR) to reduce 44 attributes to the 20 most significant, improving model accuracy and interpretability
- Tuned and deployed a CatBoost model (99.56% accuracy) as a production Flask REST API, using SHAP visualizations to communicate feature-level insights
- **Tech Stack:** Python · Pandas/NumPy · Scikit-learn · XGBoost / LightGBM / CatBoost · SHAP · Optuna · SMOTE

BigMart Sales Data Pipeline | [GitHub](#)

- Designed a normalized MySQL schema (products, outlets, sales) modeling the many-to-many item-outlet relationship across 8,523 sales records
- Built a Python/Pandas pipeline handling missing values via hierarchical imputation and standardizing inconsistent labels
- Loaded cleaned data into BigQuery, ran aggregate SQL queries, and visualized results
- **Tech Stack:** Python · Pandas/NumPy · MySQL · Google BigQuery

News Bias Analyzer

- **Built a real-time Chrome extension** that detects fake news, analyzes political bias and emotional tone in news articles.
- Designed a client-server architecture connecting a React-based Chrome extension to a Python Flask backend, integrating multiple deep learning models.
- **Tech Stack:** Python · Flask · FastAPI · React · BiLSTM · RoBERTa · spaCy · NLP · Chrome Extension API

Research and Achievements

- **Conferences:**
 - * “Polycystic Ovarian Disease (PCOD) Prediction Using Predictive Modeling” — Research Paper presented at **RCAAI 2025** (International Conference).
 - * “A Real-Time Browser-Based System for News Credibility, Bias, and Emotional Tone Analysis Using Machine Learning” — Research Paper presented at **IEEE I3CTCON 2026**.
- **Certifications:** Python Essentials by Cisco | PostgreSQL for Everybody Specialization by University of Michigan | GenAI Powered Data Analytics Job Simulation by Tata | AI Foundations by Oracle

Skills

Programming Languages: Python, SQL, Java, C

Libraries & Frameworks: Pandas, NumPy, Scikit-learn

AI and Data Engineering: Data Structures & Algorithms (Intermediate), Database Design, Semi-Structured Data Parsing, Object-Oriented Design (OOD), ETL Pipelines

Databases & Cloud: MySQL, PostgreSQL, Google BigQuery

Tools & Technologies: Git, GitHub, PostgreSQL, Vercel, VS Code, Jupyter Notebook